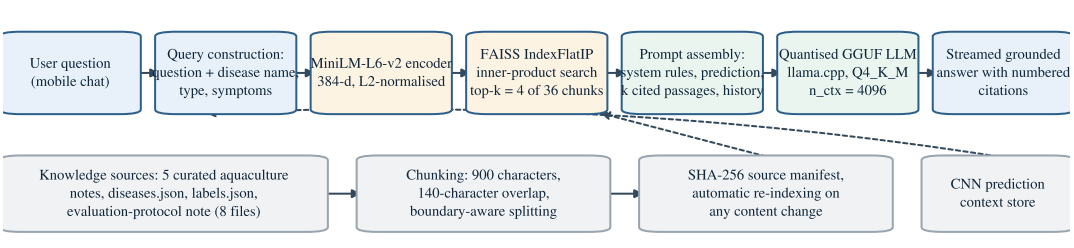
User question (mobile chat)
Query construction: question + disease name, type, symptoms
MiniLM-L6-v2 encoder 384-d, L2-normalised
FAISS IndexFlatIP inner-product search top-k = 4 of 36 chunks
Prompt assembly: system rules, prediction, k cited passages, history
Quantised GGUF LLM llama.cpp, Q4_K_M n_ctx = 4096
Streamed grounded answer with numbered citations
Knowledge sources: 5 curated aquaculture notes, diseases.json, labels.json, evaluation-protocol note (8 files)
Chunking: 900 characters, 140-character overlap, boundary-aware splitting
SHA-256 source manifest, automatic re-indexing on any content change
CNN prediction context store